Guidelines for Collaborative Task between Humanoid Robot and Quadruped Robot

1. Objectives

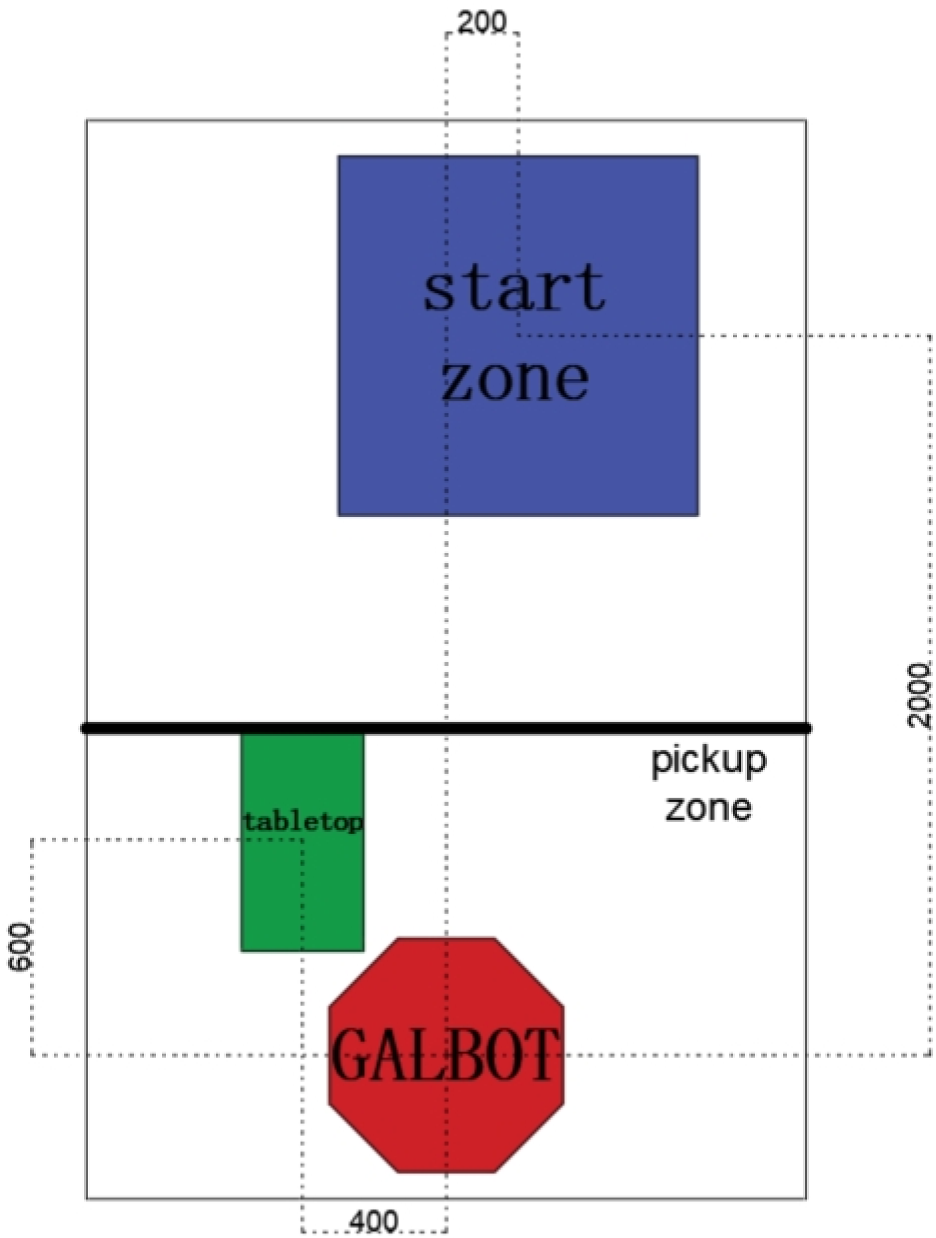
This experiment aims to evaluate the following student competencies:

- **Multi-robot collaboration**: Coordinated control between a humanoid robot (Galbot) and a quadruped robot (Go2).
- **Visual localization and motion planning**: Quadruped robot localization based on AprilTag, 6D pose estimation of objects, and robotic grasping.
- **System integration and robustness**: Code migration capability from simulation to real-world hardware, addressing real-world challenges such as sensor noise and environmental interference.
- **Safety protocols**: Ensuring no equipment damage or personnel hazards during the experiment.

2. Task Description

Task workflow:

a. **Quadruped robot localization**: Use the humanoid robot's head-mounted camera to detect the AprilTag on the robot dog's back and determine its position.
b. **Quadruped robot movement**: Send commands to control the robot dog to move collision-free to a designated point (pickup location) beside the humanoid robot.
c. **Object grasping**: Use the humanoid robot's wrist-mounted camera to identify the 6D pose of a specified object on the table, then plan and execute a robotic arm trajectory to grasp it.
d. **Object placement**: Place the object into the plastic basket on the robot dog's back.
e. **Object transport**: The robot dog carries the object back to the starting point to complete the delivery task.



3. Equipment and Environment

3.1. Hardware Configuration

- **Humanoid robot**: Galbot, equipped with a head-mounted depth camera (e.g., RealSense D435I or D436) and a wrist-mounted depth camera (D405).
- **Quadruped robot**: Go2, fitted with a pre-configured AprilTag (ID preset) and a back-mounted plastic basket.
- **Experiment scenario**:
  - Starting zone (initial position of the robot dog).
  - Pickup zone (table with target objects).
  - Obstacles (optional, to test path-planning capabilities).

3.2 Software Requirements (for reference)

- **WebSocket**: For multi-robot communication.
- **Vision algorithms**: AprilTag detection, 6D object pose estimation.
- **Motion planning**: MoveIt (robotic arm), machine dog SDK (e.g., Unitree Go2).

4. Rules and Constraints

4.1 Experiment Procedure

- **Preparation phase (5 minutes)**:
  - Verify device connections and calibrate sensors if possible.
  - Submit code and explain key algorithms (e.g., localization, grasping strategy).
- **Execution phase (10–20 minutes)**:
  - Each team has up to 2 attempts, with the highest score recorded.
  - **Timeouts** (**10 min** for each trial) result in immediate termination, with partial scores recorded.
- **Scoring phase**: On-site validation by judges (video/sensor logs).

4.2 Safety Protocols

- **Emergency stop**: Terminate the experiment immediately under the following conditions:
  - Severe collisions involving the robot dog or robotic arm (judged manually).
  - Code-induced device malfunctions (e.g., motor overheating, joint limits exceeded).
  - Failure to halt manual interventions promptly (e.g., unhandled object drops).
- **Penalty**: A score of 0 for the attempt if code causes major incidents (e.g., equipment damage).

4.3 Environmental Interference (Optional)

- Judges may introduce random disturbances (e.g., relocating objects, adjusting lighting) to test robustness.

5. Scoring Criteria (Total: 8 points)

5.1 Quadruped Robot Movement (2.5 points)

Item	Scoring Details	Points
AprilTag detection	Stable output of ID and pose (at least once)	0.5
Path planning	Collision-free movement to pickup point (0.5-point deduction per collision)	1.0
Localization accuracy	Stops within 1.5m radius of robot; returns within 1m radius of start	1.0

5.2 Humanoid Robot Operation (4.5 points)

Item	Scoring Details	Points
6D object pose estimation (grasping pose detection)	Gripper reaches reasonable grasping pose (will be judged manually)	2.0
Grasping success rate	Holds object for 2 seconds without dropping	1.0
Placing Precision	Object placed in basket without falling out	0.5
Motion planning safety	No robotic arm errors or collisions	1.0

5.3 System Integration (1 point)

Item	Scoring Details	Points
Task completion	Object delivered to the starting point without dropping (completing the task)	0.5

Process coherence	Fully automated workflow without manual interruption	0.5

Bonus Points (2 points)

- Different Grasping tests (1 point each)

6. Submission Materials

1. **Code**: Executable code.
2. **Documentation**: Algorithm explanations (e.g., localization, grasping strategy), analysis of simulation vs. real-world differences, and mitigation logic.
3. **Logs**: Sensor data (AprilTag detection, robotic arm trajectories, etc.).

7. Notes

- **Real-world discrepancies**: Perfect localization/grasping in simulations may fail on hardware due to noise; implement fault-tolerant mechanisms.
- **Collaboration interfaces**: Communication protocols (e.g., ROS Topic naming) between robots must be agreed upon in advance.
- **Contingency plans**: Redundant strategies (e.g., reattempting failed grasps) are recommended.

8. Robot control guide

Grading table	

Team No.____ Team Members:________________________________ _______		Final Score: _____	
Criteria	Description	Max pt	Scored pt
Part I	**Quadruped Robot Movement**		
AprilTag Detection	Stable output of ID and pose (at least once)	0.5	
Path Planning	Collision-free movement to pickup point (-0.5 per collision)	1.0	
Localization accuracy	Stops within 1.5m of robot, and returns within 1m radius of start	1.0	
Part II	**Humanoid Robot Operation**		
Grasping Pose Detection	Gripper reaches reasonable grasping pose	2.0	
Grasping success rate	Holds object for 2 seconds without dropping	1.0	
Placing precision	Object placed in basket without falling out	0.5	
Motion planning safety	No collisions or arm errors when moving arms	1.0	
Part III	**System Integration**		
Task completion	Complete the full task with object carried back to starting zone	0.5	
Process coherence	No manual interruption, no errors or collisions	0.5	
Part Bonus	**Additional Grasping Tests**		
Test 1	Complete the grasp without dropping	1.0	
Test 2	Complete the grasp without dropping	1.0	
Confirmation	**I confirm the assessment result and have no objections.** **Signature:** _______________________________________		